

Athindran Ramesh Kumar

r.athindran@gmail.com

Webpage: <https://athindran.github.io/>

OUTLINE

Domain expertise in controls, machine learning, and robotics. Ph.D research work on safety certification using optimization, learning, and control. TA for several courses on machine learning and data science.

EDUCATION

Princeton University

NJ, USA

MA + PhD, Electrical and Computer Engineering

Sep.2018 - Sep 2023

Advisor: Prof. Peter J. Ramadge

GPA : 3.93/4.0

- Key Courses: Machine learning and Pattern Recognition, Modern Control, Safe Robotics, Theoretical Machine Learning, Optimization for Machine Learning, Reinforcement Learning.
- M.A. degree in Electrical Engineering awarded.
- Ph.D. dissertation under evaluation currently - to be awarded Sep 2026.

University of Illinois at Urbana-Champaign

Illinois, USA

MS (fully funded by Dept.), Electrical and Computer Engineering

Aug. 2013 - Aug 2015

Advisor: Prof. Grace Gao

GPA : 3.95/4.0

Indian Institute of Technology, Madras

Chennai, India

B.Tech, Electrical Engineering

Aug. 2009 - July 2013

Advisor: Prof. Radhakrishna Ganti

GPA : 9.27/10.0

PROFESSIONAL EXPERIENCE

• Aurora Innovation

Pittsburgh, PA

Software Engineer II - Behavior Planning and Control

October 2023 - present

- Contributed to state-of-the-art ML models for motion planning and scene understanding
- Experience working on safety-critical robotics software that implements control algorithms on a self-driving truck while ensuring fail-safe testing

• Aurora Innovation

Pittsburgh, PA

Software Intern - Controls

May - Aug 2022

- Analysis and deployment of improvements to longitudinal control of autonomous trucks

• Nokia Bell Labs

Murray Hill, NJ

Research Intern

Jun - Aug 2021

- Reinforcement learning algorithms for a multi-link robotic arm in simulation
- Sample-efficient reinforcement learning with expert demonstrations
- Adapting efficiently to re-configurable robotic arms using sequence neural networks

• Center of Excellence in Wireless Technology

Chennai, India

Research Engineer

Apr 2016 - June 2018

- Path loss modeling with GIS satellite imagery from geo-platform of ISRO
- Developed frequency planning methods in a communication system that have been patented by the organization

• IIT Madras

Chennai, India

Project Associate

Nov 2015 - Mar 2016, Jul 2017 - Jul 2018

- Worked on a proposal seeking funding for the 5G mmWave cellular project at IIT Madras
- Efficient deep learning for object detection in videos using inference-time schemes for acceleration

• Google Inc.

Mountain View, CA

Software Intern - Street View

May - Aug 2014

- Implemented ambiguity resolution algorithms in Python on GPS carrier phase data obtained from receivers installed on Street View cars to achieve sub-meter accurate positioning

SELECT PUBLICATIONS

Journal Papers

- **A.R. Kumar**, K. -C. Hsu, P. J. Ramadge and J. F. Fisac, “Fast, Smooth, and Safe: Implicit Control Barrier Functions through Reach-Avoid Differential Dynamic Programming,” in IEEE Control Systems Letters, doi: 10.1109/LCSYS.2023.3292132.
- Heng, Liang, A.R. Kumar, and Grace Gao. “Private proximity detection using partial GPS information.” IEEE Transactions on Aerospace and Electronic Systems 52.6 (2016): 2873-2885.

Conference and Workshop Papers

- **S. Liu, A.R. Kumar**, Jaime F. Fisac, Ryan P. Adams, Peter J. Ramadge. “ProBF: Probabilistic Safety Certificates with Barrier Functions.” Presented at SafeRL workshop at NeurIPS 2021.
- **A.R. Kumar** and Peter J. Ramadge. “Learning to Control Using a Convex Combination of Controllers.” 2021 American Control Conference (ACC). IEEE, 2021.
- **A.R. Kumar** and Peter J. Ramadge, 2021, March. DiffLoop: “Tuning PID controllers by differentiating through the feedback loop.” In 2021 55th Annual Conference on Information Sciences and Systems (CISS) (pp. 1-6). IEEE.
- T. H. Fan, A. R. Kumar and P. J. Ramadge. Safety Control for Prime Focus Spectrograph. In 2022 56th Annual Conference on Information Sciences and Systems (CISS) (pp. 269-274). IEEE.
- **A.R. Kumar**, Balaraman Ravindran, and Anand Raghunathan. “Pack and detect: Fast object detection in videos using region-of-interest packing.” Proceedings of the ACM India Joint International Conference on Data Science and Management of Data. 2019.
- **A.R. Kumar**, Liang Heng, and Grace X. Gao. “GPS privacy: Enabling proximity-based services while keeping GPS location private.” Proceedings of the 27th International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GNSS+ 2013),(Tampa, FL). 2014.

Patents

- **Athindran R**, Navinnath P, Klutto Milleth, Bhaskar Ramamurthi, “Frequency Assignment for SINR and Throughput Management in Battlefield Communication”, India Patent granted 27th June 2024.

SELECT ACADEMIC ACHIEVEMENTS

- Awarded full-tuition waiver and stipend for MS degree program at University of Illinois, Urbana-Champaign
- Received first-year fellowship at Princeton University for PhD program
- Outstanding merit in Mathematics from Srinivas Ramanujan academy of Maths talent awarded in 2008
- Ranked 294 out of 1,000,000 students in AIEEE and 1561 out of 800,000 students in JEE

ACADEMIC SERVICE

Teaching Experience

- Three-time TA for ECE 435-535 (Machine learning course with strong math foundations)
- Assistantship in Teaching for 11 semesters

Reviewing Service

- Conferences: ICLR (2021, 2023, 2024), NeurIPS (2022-2024), ICML (2023-2025), CISS 2022, IJCAI 2024, IJCAI 2026, CDC 2026
- Journals: IEEE Transactions on Control Systems Technology, IEEE Robotics and Automation Letters
- **Top reviewer for NeurIPS 2023**
- **Recognition of reviewing contribution within the PPI - Bronze Tier for IJCAI 2026**

SELECT PROJECTS

–Optimization and Learning methods for Safety-Critical Control–

Princeton University, NJ

Guide: Prof. Peter Ramadge

Jul 2019 - Aug 2023

- Safety certification for autonomous control systems
- Learning residual dynamics using probabilistic models

–Efficient Deep Learning for Videos–

IIT Madras, Chennai

Guide: Prof. B. Ravindran and Prof. Anand Raghunathan (Purdue University)

Jul 2017 - Jul 2018

- Novel inference time method to accelerate object detection in videos
- Published **ACM India Joint International Conference on Data Science and Management of Data 2019**

PROGRAMMING SKILLS

- Python - PyTorch - JAX - C++ - Matlab